

Rapid Peer Education of a Computational NE Skill Suite

ANS Annual Conference, Hollywood, FL

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June 30, 2011





Outline

1 Motivation

State of Nuclear Engineering
Requisite Skill Suite
University Curriculum

2 Organization

BiWeekly Meetings
Listhost
Wiki

3 Bootcamps

Audience
Topics
Technique
Feedback



State of Nuclear Engineering

Applications

Lattice Physics

Neutronics

Thermal Hydraulics

Fuel Performance

Criticality Safety

Depletion

Numerical Methods

Advanced Monte Carlo

Turbulent Flow

Sensitivity Analysis

Risk Assessment

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Figure: Image courtesy of Argonne National Laboratory.

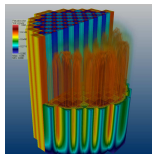


Figure: Image courtesy of ORNL.

Requisite Skill Suite



Essential Skills

Parallelization

Databases

Unit Testing

Version Control

Fortran/C++/Python

Statistics

Advanced Visualization

Development Workflow

Documentation

Debugging

Unix/Linux

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Figure: Image courtesy of Argonne National Laboratory.

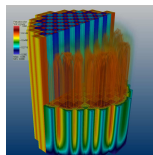


Figure: Image courtesy of ORNL.

University Curriculum is Limited



CS 367	Introduction to Data Structures
CS 412	Introduction to Numerical Methods
EP/EMA 471	Engineering Problem Solving II
EP/EMA/NE 476	Computational Engineering

Table: Undergraduates must take one course from the list above.

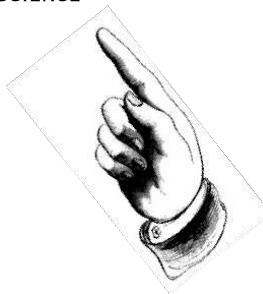
NE476:

Course Description: Basic tools of professional scientific computation for UNIX environments are taught. Programming skills in a compiled language are developed through engineering examples. Applications reinforce engineering problem-solving skills first examined in introductory courses, while motivating progressively more advanced computational methods.

Department Scope

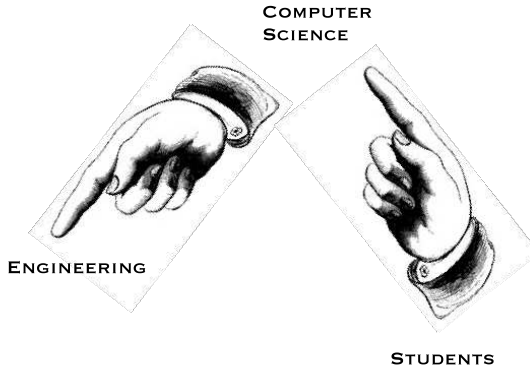


**COMPUTER
SCIENCE**

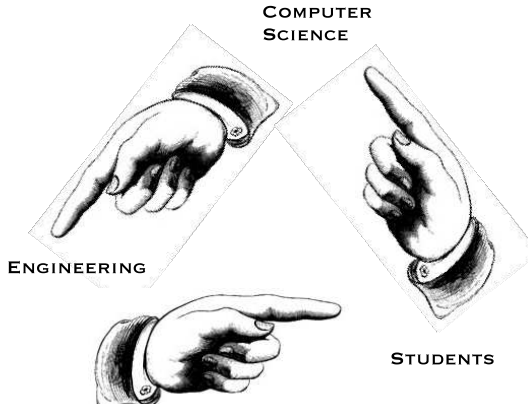


STUDENTS

Department Scope



Department Scope



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TIRW

BiWeekly Meetings



Autotools
Backup
Build Systems
Cmake
Debugging
Doxygen
GpGpu
LaTeX
MachineLearning
Parallelization with MPI
PlayStation3
PyCon2010
Sahana Haiti Relief
The Shell
Unit Testing
Visualization Course

Automated Documentation
Beamer
C++0x
Condor
In Defense of Fortran
FileSystem
GprofDdd
LibraryDesign
Matplotlib
Parallelization with Python
PrettyPrinters
PythonCinterface
SavingScholarship
ShowAndTell
Version Control
Web Programming

Awesome Sauce
Bert Computing Cluster
C++ And Fortran
Cython
Distributed Version Control
FuturePlanning
Large Sciency Documents
LightningTalks
OpenAccess
PerlDataLanguage
Processes
R
Scripting
UnixAccess
Visualization
XML



Audience

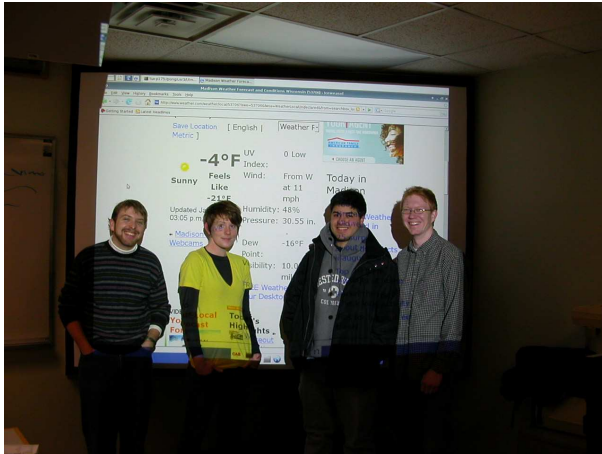


Figure: Hacker Within regulars develop a community.

Listhost



An interdisciplinary, peer driven listhost of approximately 130 students gives students a quick way to ask questions of an experienced but casual group.



Hacker Within

Python question

[Options](#)

★ 10 messages - [Expand all](#) - [Report discussion as spam](#)

- Emily B. Sessa** Hi, Wolfman sent me to seek advice from this group... I've got a very beginner-level question about Python - I'm a grad student and just need to ru... May 21, 11:42 pm
- Matt Terry** Wolfman is a wise one :) I'm going to guess that numpy/scipy are installed correctly, but are installed for a version than the default python. From the co... May 22, 1:32 am
- Anthony Scopatz** Also, the command 'python -V' will tell you which version of Python you have installed, so you don't have to worry about guessing. On Sun, May ... May 22, 3:23 am
- Mr. Puneet Kishor** \$ which python will tell you the first python instance in your path. You can have multiple python versions and call them directly as '/usr/local/bin/... May 22, 9:24 am
- esessa** Hi all, Thank you so much! This has already helped! Taking Matt's advice, up top, and just typing python2.6, I get this, which is what I was hoping for... call... May 22, 12:43 pm
- Anthony Scopatz** - Hide quoted text - Show quoted text - On Sun, May 22, 2011 at 11:43 AM, esessa <ese...@wisc.edu> wrote: > Hi all, > Thank you so much! Thi... May 22, 1:01 pm



hackerwithin.org

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Bootcamps



Topic	Year	Days	$\frac{hr}{day}$	Attendees
Unix	2009	4	2	18
C++	2009	4	2	30
Python	2010	3	4	82
Software Carpentry	2011	3	4	79
MSU	2011	2	5	40

Table: Previous Bootcamps have been conducted in two condensed formats. Feedback surveys from early bootcamps led to a three rather than four day structure.

Programming Experience

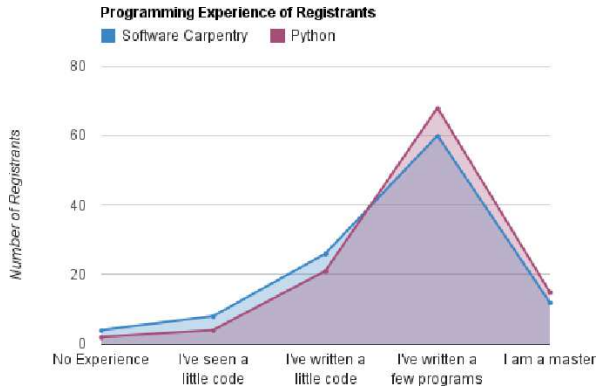


Figure: The participants reported varying levels of experience.

University Standing

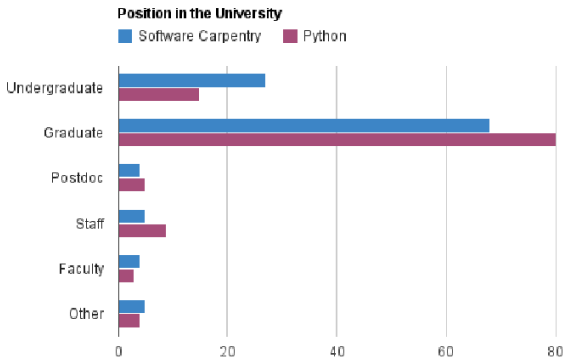


Figure: The participants reported varying levels of education.



Disciplines

Departments

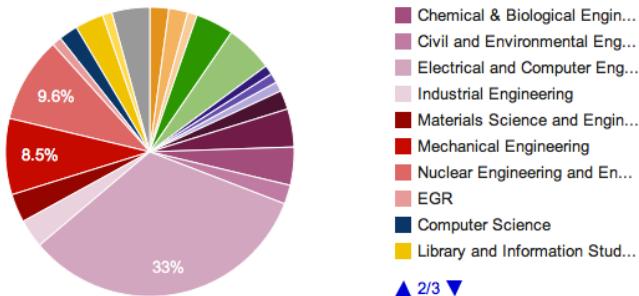


Figure: The participants come from various backgrounds. Here is the distribution from the Software Carpentry bootcamp.



Bootcamp Topics

Unix	C++	Python	S.C.
Shell Environment RegEx Make Autotools	basics conditionals loops functions multi-file progs inheritance pointers data structures templates exceptionss	installation data types modules numpy/scipy matplotlib django pytables swig/f2py pytables wxPython GUIs	Shell Version Control Text Editors Databases Web Programming Auto. Documentation Build Systems Unit Testing Debugging

Table: The bootcamps have covered myriad topics pertinent to a fundamental computational nuclear engineering skill suite.

Virtual Machine



Before the bootcamp, students are asked to:

- Install VirtualBox
- Download a 1GB file

When the bootcamp begins, the students open their virtual machines and may immediately perform exercises and try out tools on identical ubuntu computers pre-loaded with an array of appropriate software.



Virtual Machine

For the software carpentry bootcamps the virtual machines were loaded with an array of awesome (free) software.

- python 2.6
- ipython
- scipy, numpy, matplotlib, pygame, pytables
- emacs
- vim
- nano, pico, nedit
- g++
- gfortran
- cmake
- doxygen
- mercurial, git, bzr, svn



Example Driven

Source code is provided by a shared repository. Each student checks out their own version and makes local changes in sync with the class.

C++ example with CMake

CMake is a C/C++ build system (with some additional support for Fortran, Java, and Python extensions). The website description:

Welcome to CMake, the cross-platform, open-source build system. CMake is a family of tools designed to build, test and package software. CMake is used to control the software compilation process using simple platform and compiler independent configuration files. CMake generates native makefiles and workspaces that can be used in the compiler environment of your choice.

Get the example code

Our first order of business is to get you access to these notes and other course materials. We've placed them in a repository on our [Google Code site](#). However, instead of having you visit the site and download them manually, we will download the latest version with version control.

You may check out a read only or a committing copy of the repository. To check out a read only copy of today's session::

```
mkdir build_systems
cd build_systems
svn co http://hacker-within.googlecode.com/svn/trunk/sc/documentation/src cxx-example
```

Build the example

CMake is designed to keep files that result from the build process separate from the source code directory. This makes it easier to keep the source directory under version control, and it makes it easy to start a fresh configuration and build with a simple `rm -rf` on the build directory.

Let's make a build directory

```
mkdir cxx-example_build
cd cxx-example_build
```

CMake has a Turing complete scripting language with scripting directives designed specifically for a C/C++ build system. These

Peer Taught

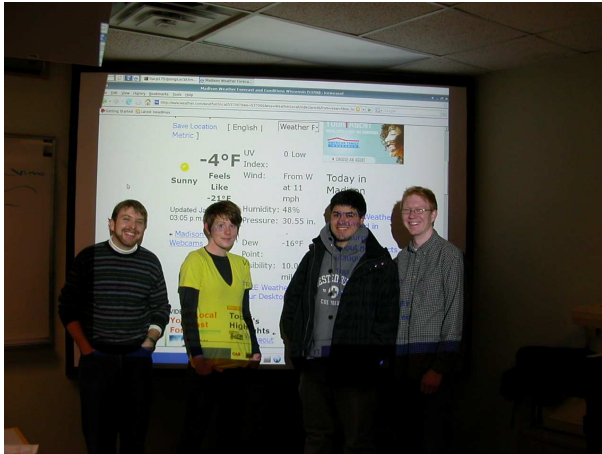
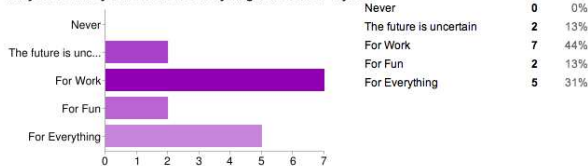


Figure: Hacker Within regulars learn by teaching and are not terribly intimidating.

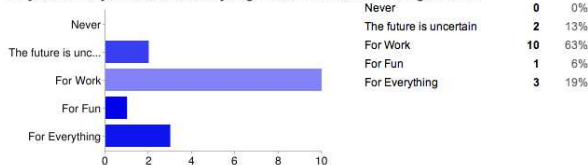
Feedback : Python Bootcamp I



Will you use what you heard about for anything in the future? - Python



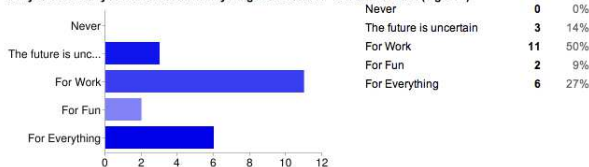
Will you use what you heard about for anything in the future? - hackerwithin.org wiki notes



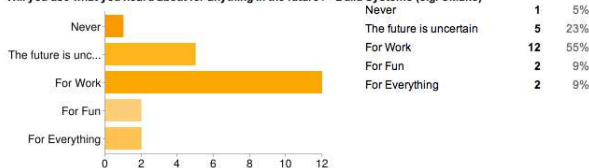
Feedback : Software Carpentry Bootcamp



Will you use what you heard about for anything in the future? - Version Control (e.g. svn)



Will you use what you heard about for anything in the future? - Build Systems (e.g. CMake)



Lessons



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Lessons



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- and best practices are absent from curriculum.
- Students appreciate a swift introduction to software tools
- particularly one that is hands on
- and example driven.
- A community is helpful
- and can drive itself.



References I

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